MATH423: Introduction to Computational Topology

Triangulations, Classification of Surfaces and Abstract Simplicial Complexes

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Week 2: Surfaces and their Triangulations (continued)

2.4 Triangulations and the Euler Characteristic

Polygonal presentations are very economical: a closed surface is described by a single polygon and a word. For computation, however, they have two drawbacks. First, the gluing is encoded globally: to understand what the surface looks like near one corner of the polygon, we had to trace the identifications through the whole word. Second, real data never arrives in this form. A surface scanned by a 3D scanner, a surface in a computer graphics program, or a surface reconstructed from sample points is given by many small pieces, each of which only knows its immediate neighbors.

Computational topology therefore describes spaces by gluing together many copies of the simplest possible building block. In dimension two this block is the triangle, and the resulting description is a triangulation. It has three features that the rest of the course relies on.

- **Uniformity.** Every piece has the same shape, so the same few lines of code handle every piece.
- **Locality.** The structure near a point can be read off from the few triangles touching it, without looking at the rest of the surface.
- **Finiteness.** The whole surface is a finite list of triangles, each of which is a list of three vertices. This is something a computer can store exactly.

Definition 2.8. A **triangulation** of a compact surface S consists of a finite number of triangles covering S such that the intersection of any two triangles is either empty, a single common vertex, or a single common edge.

Equivalently, a triangulation is a simplicial complex K such that its polyhedron $|K|$ is homeomorphic to S. By a theorem of Radó (1925), every compact surface admits a triangulation.

Example 2.9. The intersection condition is what makes a triangle determined by its three vertices. Take two triangles and glue them along their whole boundaries, vertex

to vertex and edge to edge. The result is a sphere (a pillow with three corners), and counting gives $V = 3$, $E = 3$, $F = 2$. But this is *not* a triangulation in the sense above: the two triangles meet in all three edges, and they have the same three vertices, so the list of vertex triples would contain $\{1, 2, 3\}$ twice and could not tell the two triangles apart. In a triangulation, a triangle is uniquely specified by its set of vertices. This is the reason a triangulated surface can be stored simply as a list of triples. For instance, the boundary of the tetrahedron is the list

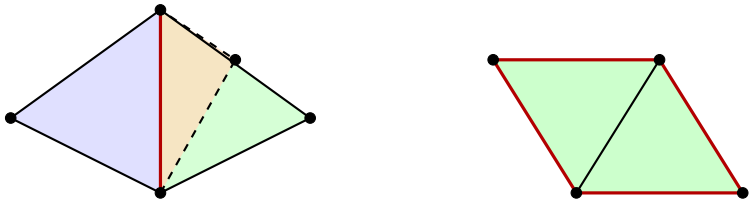
$$\{1, 2, 3\}, \quad \{1, 2, 4\}, \quad \{1, 3, 4\}, \quad \{2, 3, 4\},$$

and from this list alone one recovers its 6 edges, as the 2-element subsets of the triples, and its 4 vertices.

Remark 2.10. For surfaces nothing is lost by working with triangulations, since every compact surface has one. The same is true in dimension 3 (Moise, 1952). It fails in higher dimensions: there are topological 4-manifolds, such as Freedman's E_8-manifold, that admit no triangulation, and Manolescu (2016) showed that non-triangulable manifolds exist in every dimension $n \geq 5$. Such spaces are invisible to any method that starts from a simplicial complex. In practice this causes no difficulty, because the spaces arising from data are built as complexes in the first place.

Lemma 2.11. *Let K be a triangulation of a closed surface. Then every edge of K is a face of exactly two triangles, and every vertex and every edge of K is a face of at least one triangle.*

Proof sketch. Let e be an edge contained in exactly k triangles, and let x be a point in the interior of e. If $k = 0$, then x has a neighborhood homeomorphic to an open interval, which is not homeomorphic to an open disk (removing a point disconnects an interval but not a disk). If $k \geq 1$, a small neighborhood of x consists of k half-disks glued along their common diameter, like a book with k pages. For $k = 1$ this is a half-disk and x would be a boundary point. For $k = 2$ the two half-disks form a disk. For $k \geq 3$ the open book is not homeomorphic to an open disk; this is intuitively clear, and a rigorous proof uses the local homology groups of Week 5 (at x the book has local homology of rank $k - 1$, a disk has rank 1). Hence $k = 2$. A vertex or an edge contained in no triangle would similarly have a neighborhood that is a graph, not a disk. □



(a) an edge in three triangles (b) a disk: edges in one triangle

Figure 7: The two ways the Lemma can fail. In (a) no point inside the common edge has a disk neighborhood. In (b) each of the four outer edges lies in only one triangle, so the surface has boundary and is not closed. In mesh processing these are called non-manifold edges and holes, and software for 3D printing searches for exactly these two defects.

Counting the pairs (edge, triangle containing it) in two ways, the lemma gives the useful relation

$$3F = 2E$$

for every triangulation of a closed surface with E edges and F triangles: each triangle has 3 edges, and each edge is counted by exactly 2 triangles.

Given a triangulated surface, we can count its structural components to compute a fundamental topological invariant.

Definition 2.12. Let a triangulated surface have V vertices, E edges, and F faces (triangles). The **Euler characteristic** of the surface is defined as:

$$\chi = V - E + F$$

The Euler characteristic is our first example of a pattern that runs through the whole course: a *global* property of a space computed by purely *local* counting. To compute χ, a program reads the list of triangles once, collects the distinct vertices and edges, and counts. It does not need to know what the surface looks like, how it sits in space, or even which surface it is.

Theorem 2.13. *The Euler characteristic is a topological invariant. It does not depend on the choice of triangulation.*

We will prove this in Week 5, where χ will be expressed as the alternating sum of the ranks of the homology groups. More generally, the same formula $V - E + F$ applies to any decomposition of a surface into vertices, edges and polygonal faces whose interiors are open disks; this is what we used for polygonal presentations above.

Why the alternating sum?

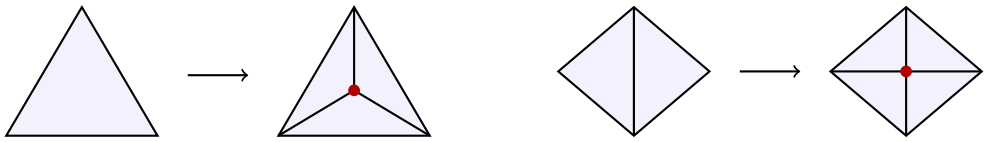
The individual numbers V, E and F are *not* invariants: refining a triangulation increases all of them. The point of the alternating sum is that the increases cancel. There are two basic ways to refine a triangulation of a surface, shown in Figure 8.

1. **Subdividing a triangle.** Add a new vertex inside a triangle and join it to the three corners. One triangle is replaced by three, so V increases by 1, E by 3 and F by 2.

2. **Subdividing an edge.** Add a new vertex in the middle of an edge and join it to the opposite corners of the two triangles containing that edge. The edge is split into two and two new edges appear, while two triangles are replaced by four, so again V increases by 1, E by 3 and F by 2.

In both cases the change in χ is $1 - 3 + 2 = 0$. A classical theorem, going back to Radó, Alexander and Newman, says that any two triangulations of the same closed surface can be connected by a finite sequence of these two moves and their inverses. This reduces the theorem to the computation above. The proof in Week 5 is different, and it works in every dimension.

Example 2.14. Consider the boundary of a tetrahedron, which is homeomorphic to the sphere S^2. We have $V = 4$, $E = 6$, and $F = 4$, yielding $\chi = 4 - 6 + 4 = 2$. If we instead look at a cube (also homeomorphic to S^2, though its faces are not triangles, the formula still applies for cell complexes), we have $V = 8$, $E = 12$, $F = 6$, which gives $\chi = 8 - 12 + 6 = 2$. The boundaries of the octahedron ($V = 6$, $E = 12$, $F = 8$) and the icosahedron ($V = 12$, $E = 30$, $F = 20$) are further triangulations of S^2, and both again



(1) subdividing a triangle (2) subdividing an edge

Figure 8: The two basic refinements of a triangulated surface. Each adds 1 vertex, 3 edges and 2 triangles, so the Euler characteristic does not change.

give $\chi = 2$. The tetrahedron and the octahedron are related by the moves above: label the tetrahedron $\{1, 2, 3, 4\}$, subdivide the edge $\{1, 2\}$ and then the edge $\{3, 4\}$. Each of the six vertices now has degree 4, and the result is the octahedron, with $\chi = 2$ at every step.

For a closed surface, the relations $3F = 2E$ and $V - E + F = \chi$ can be solved for E and F:

$$E = 3(V - \chi), \qquad F = 2(V - \chi).$$

So once the surface is known, the number of vertices determines the size of the whole triangulation. For example, every triangulation of S^2 with V vertices has exactly $3V - 6$ edges and $2V - 4$ triangles, and every triangulated closed surface has an even number of triangles. Relations of this kind are used to estimate memory: a mesh of a closed surface with a million vertices has about two million triangles and three million edges, whatever the surface is, because χ is negligible compared to V.

Example 2.15. To triangulate the torus, divide the square $I \times I$ into a 3×3 grid of small squares and cut each small square along a diagonal. After the identifications $(0, y) \sim (1, y)$ and $(x, 0) \sim (x, 1)$, the 16 grid points become 9 vertices. Each of the 9 small squares contributes 2 triangles, so $F = 18$, and $E = 3F/2 = 27$. Therefore

$$\chi(T^2) = 9 - 27 + 18 = 0.$$

A 2×2 grid does *not* work: after the identifications it has only 4 vertices, and then two distinct triangles would share the same three vertices, which is not allowed in a simplicial complex.

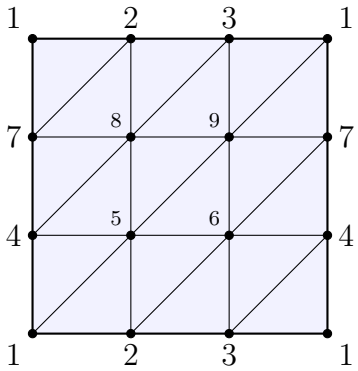


Figure 9: A triangulation of the torus with 9 vertices, 27 edges and 18 triangles. Vertices with the same label are identified.

Example 2.16. How small can a triangulation of the torus be? Since $\chi(T^2) = 0$, a triangulation with V vertices has $E = 3V$ edges. On the other hand, distinct edges have distinct pairs of endpoints, so $E \leq \binom{V}{2} = \frac{V(V-1)}{2}$. Hence $3V \leq \frac{V(V-1)}{2}$, that is, $V \geq 7$. The bound is attained: there is a triangulation of the torus with 7 vertices, 21 edges and 14 triangles, already known to Möbius, in which every pair of vertices is joined by an edge. The same counting argument applies to any closed surface, but the bound it gives is not always attained: it also gives $V \geq 7$ for the Klein bottle, whose smallest triangulation has 8 vertices (Franklin, 1934). Minimal triangulations matter in practice, since the cost of every algorithm in this course grows with the size of the complex.

We can calculate the Euler characteristic of connected sums using the formula:

$$\chi(S_1 \# S_2) = \chi(S_1) + \chi(S_2) - 2$$

Proof sketch: Choose triangulations of S_1 and S_2 and form the connected sum by removing the interior of one triangle from each and gluing the two boundary triangles together, vertex to vertex and edge to edge. Removing the interiors drops the face count by 1 in each surface, giving $(\chi(S_1) - 1) + (\chi(S_2) - 1)$ for the disjoint union. The gluing then identifies 3 pairs of vertices and 3 pairs of edges, so V decreases by 3 and E decreases by 3, and these changes cancel in $V - E + F$. Thus $\chi(S_1 \# S_2) = \chi(S_1) + \chi(S_2) - 2$.

For example, since $\chi(T^2) = 0$, the genus-g orientable surface formed by the connected sum of g tori has an Euler characteristic of $2 - 2g$. Similarly, since $\chi(\mathbb{R}P^2) = 1$, the non-orientable surface formed by the connected sum of k projective planes has an Euler characteristic of $2 - k$. Both agree with the computations from polygonal words above.

Remark 2.17. The Euler characteristic also makes sense for compact surfaces with boundary. The closed disk (a single triangle) has $\chi = 3 - 3 + 1 = 1$. Since removing an open disk from a closed surface lowers χ by 1 (as in the proof above), the cylinder $= S^2$ minus two disks has $\chi = 0$, and the Möbius band $= \mathbb{R}P^2$ minus a disk also has $\chi = 0$. Thus the cylinder and the Möbius band have the same Euler characteristic; they are distinguished by orientability (or by the number of boundary circles).

2.5 Classification of Closed Surfaces

The classification theorem provides a complete list of all closed (compact and without boundary) connected surfaces up to homeomorphism.

Theorem 2.18 (Classification of Closed Surfaces)**.** *Any closed, connected surface is homeomorphic to exactly one of the following:*

1. *The sphere S^2.*
2. *The connected sum of $g \geq 1$ tori (orientable surface of genus g). Its Euler characteristic is $2 - 2g$.*
3. *The connected sum of $k \geq 1$ projective planes (non-orientable surface). Its Euler characteristic is $2 - k$.*

Corollary 2.19. *Two closed connected surfaces are homeomorphic if and only if they have the same orientability and the same Euler characteristic. In particular, a closed connected surface with odd Euler characteristic is non-orientable.*

Remark 2.20. Mixed connected sums do not produce new surfaces. By Dyck's theorem,

$$T^2\#\mathbb{R}P^2 \cong \mathbb{R}P^2\#\mathbb{R}P^2\#\mathbb{R}P^2.$$

Both sides are non-orientable with $\chi = -1$, consistent with the corollary. Consequently, a connected sum of g tori and $k \geq 1$ projective planes is homeomorphic to the connected sum of $2g + k$ projective planes.

Example 2.21. Identify the surface given by the hexagon word $abca^{-1}b^{-1}c^{-1}$. Every letter appears once with each sign, so the surface is orientable. Labelling the corners $P_0, \ldots, P_5$, the edges are $a : P_0 \to P_1$, $b : P_1 \to P_2$, $c : P_2 \to P_3$, $a : P_4 \to P_3$, $b : P_5 \to P_4$, $c : P_0 \to P_5$. The identifications are

$$\underbrace{P_0 \sim P_4,\ P_1 \sim P_3}_{a}, \qquad \underbrace{P_1 \sim P_5,\ P_2 \sim P_4}_{b}, \qquad \underbrace{P_2 \sim P_0,\ P_3 \sim P_5}_{c},$$

giving two vertex classes $\{P_0, P_2, P_4\}$ and $\{P_1, P_3, P_5\}$. Hence $V = 2$, $E = 3$, $F = 1$ and $\chi = 0$. An orientable closed surface with $\chi = 0$ is the torus. (The hexagonal presentation of the torus is the one that tiles the plane by regular hexagons.)

Classification as an Algorithm

The corollary says that two pieces of information, the Euler characteristic and orientability, identify a closed connected surface completely. Both can be computed from a list of triangles, so the classification theorem is in fact an algorithm. We have seen how to compute χ; it remains to explain how a computer can decide orientability, since the definition (the surface contains a Möbius band) is not something one can test directly.

Definition 2.22. An **orientation** of a triangle $\{a, b, c\}$ is a choice of cyclic order of its vertices, written $[a, b, c]$; thus $[a, b, c] = [b, c, a] = [c, a, b]$, and the opposite orientation is $[a, c, b]$. The orientation $[a, b, c]$ traverses the edges $a \to b$, $b \to c$ and $c \to a$. Two triangles sharing an edge are **consistently oriented** if they traverse the common edge in opposite directions. A triangulated surface is **consistently oriented** if all its triangles are oriented so that every pair of triangles sharing an edge is consistently oriented.

Picture each orientation as a small arrow circling inside the triangle. Two neighboring triangles are consistently oriented when their arrows circle the same way as seen from one side of the surface, which forces them to run in opposite directions along the shared edge. A triangulated surface is orientable if and only if it can be consistently oriented. This gives the following procedure.

Input: a finite list of triangles forming a closed connected surface.

1. Count the distinct vertices, edges and triangles, and compute $\chi = V - E + F$.
2. Orient one triangle arbitrarily. Repeatedly take an oriented triangle and a neighbor across one of its edges. If the neighbor is not yet oriented, give it the unique orientation consistent with the oriented triangle. If it is already oriented, check consistency.
3. If a check fails, the surface is non-orientable, and it is the connected sum of $2 - \chi$ projective planes. Otherwise it is orientable of genus $(2 - \chi)/2$.

In step 2 the orientation of the first triangle forces the orientation of each of its neighbors, which forces the orientation of their neighbors, and so on; so the procedure either produces a consistent orientation or proves that none exists. This is a breadth-first search over the triangles, with each edge examined a bounded number of times, so the whole procedure takes time proportional to the size of the input. Connectedness of the input can be checked by the same search, and whether a list of triangles forms a closed surface at all can also be decided locally, by checking that each edge lies in exactly two triangles and that around each vertex the triangles fit together in a single cycle (you will prove one direction of this in the exercises).

Example 2.23. For the 9-vertex torus of the figure above, orient every triangle counterclockwise as drawn in the square. Two triangles sharing an edge inside the square then traverse it in opposite directions. The identifications of the square are translations, which preserve the counterclockwise direction, so the same holds for edges on the boundary of the square. The orientation is consistent, and with $\chi = 0$ the algorithm reports the torus.

Example 2.24. Consider the five triangles

$$\{1,2,3\}, \quad \{2,3,4\}, \quad \{3,4,5\}, \quad \{4,5,1\}, \quad \{5,1,2\},$$

forming the strip in Figure 10. The edges $\{1,2\},\{2,3\},\{3,4\},\{4,5\},\{5,1\}$ each lie in two triangles, while the other five edges lie in one triangle only and form the single boundary circle $1-3-5-2-4-1$. Start with $[1,2,3]$, which traverses $2 \to 3$. The neighbor $\{2,3,4\}$ must traverse $3 \to 2$, so it gets $[3,2,4]$, which traverses $4 \to 3$. Continuing in the same way, $\{3,4,5\}$ gets $[3,4,5]$, then $\{4,5,1\}$ gets $[5,4,1]$, and then $\{5,1,2\}$ gets $[5,1,2]$. Now $[5,1,2]$ traverses $1 \to 2$, and so does $[1,2,3]$: the last check fails. No consistent orientation exists, and the strip is a Möbius band. Its Euler characteristic is $5-10+5=0$, as computed in the remark above.

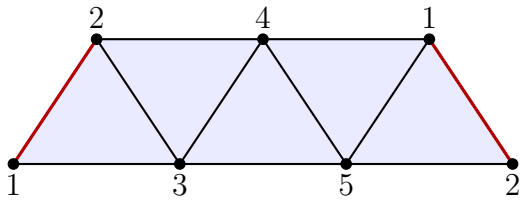


Figure 10: A triangulated Möbius band with 5 vertices. The two marked end edges are glued so that the vertex 1 at the bottom left meets the vertex 1 at the top right, which introduces the half-twist.

Every step of this procedure uses only the list of triangles, and its running time grows only linearly with the size of the list. Surfaces are therefore completely understood from the computational point of view. Nothing like this is possible in general: as discussed in Section 4 (Why Simplicial Complexes?), there is no algorithm deciding whether two triangulated 4-manifolds are homeomorphic. In higher dimensions we must settle for invariants that are computable but do not classify, and the most important of these, homology, is the subject of Week 5.

3 Week 3: Abstract and Geometric Simplicial Complexes

In the first week, we briefly introduced geometric simplices. This week, we will formalize abstract simplicial complexes, which strip away the geometry to leave only combinatorial data. This combinatorial approach is essential for representing topological spaces algorithmically. We will then introduce the standard methods for building complexes from point cloud data: the Čech and Vietoris-Rips complexes.

It helps to keep three levels of description apart:

$$\underbrace{X}_{\text{a topological space}} \xleftarrow{\cong} \underbrace{|K|}_{\text{a polyhedron in } \mathbb{R}^n} \longleftarrow \underbrace{K}_{\text{a finite list of vertex sets}}$$

The space X is what we want to understand, for example a surface or the unknown shape from which a data set was sampled. The polyhedron $|K|$ is a concrete model of X built from geometric simplices. The abstract complex K is the data structure that a computer actually stores. Last week we moved from the first level to the second; this week we move from the second to the third.

3.1 Abstract Simplicial Complexes

While a geometric simplicial complex lives in $\mathbb{R}^n$ and consists of actual geometric objects (points, line segments, solid triangles), an abstract simplicial complex is a purely combinatorial object.

Why forget the coordinates?

A geometric simplicial complex carries more information than topology needs. Its topology depends only on which sets of vertices span simplices, not on where the vertices are placed. For instance, a hollow triangle drawn with an equilateral or with a very thin shape is homeomorphic to S^1 either way. In general, if the vertices of two geometric complexes can be matched so that a set of vertices spans a simplex in one complex exactly when the matching set spans a simplex in the other, then extending the matching linearly over each simplex gives a homeomorphism between their polyhedra. Dropping the coordinates has several practical advantages.

- **Exactness.** Condition 2 in the definition of a geometric simplicial complex requires computing intersections of simplices in $\mathbb{R}^n$. With coordinates stored as floating point numbers, deciding whether two triangles meet only along an edge or overlap slightly can go wrong. The conditions in the abstract definition below are statements about finite sets of integers, and they are checked exactly.

- **No ambient space needed.** Many complexes are not naturally drawn anywhere. The real projective plane cannot be embedded in $\mathbb{R}^3$ at all, so a geometric triangulation of it must live in $\mathbb{R}^4$ or higher. The co-authorship complex of a department, or a complex built from a table of distances between DNA sequences, has no natural coordinates at all.

- **Size.** A list of vertex sets is the smallest description from which the topology can be recovered. Coordinates can be added later, if a picture is needed.

The passage from geometry to combinatorics is made by recording only the vertex sets. If K is a geometric simplicial complex with vertex set V, its **vertex scheme** is the collection of all subsets of V that span a simplex of K. For example, the vertex scheme of the boundary of the tetrahedron consists of the four triples listed in the previous week, together with their 6 subsets of size two and 4 subsets of size one. Labelling the vertices of the hollow triangle K' of Week 1 by $1, 2, 3$, its vertex scheme consists of $\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}$, and the vertex scheme of the solid triangle additionally contains $\{1, 2, 3\}$.

Any vertex scheme has two evident properties. Every vertex is a 0-simplex, so every singleton $\{v\}$ belongs to it. Every face of a simplex is again a simplex of K, so every non-empty subset of a set in the vertex scheme belongs to it as well. The second condition in the definition of a geometric complex, about intersections, has no counterpart: it is automatically encoded by the fact that simplices are recorded through their vertex sets. The definition of an abstract simplicial complex simply takes these two properties as axioms.

Definition 3.1. An **abstract simplicial complex** is a pair (V, K) where V is a set of vertices and K is a family of non-empty finite subsets of V such that:

1. For every $v \in V$, the singleton $\{v\}$ is in K.
2. If $\sigma \in K$ and $\tau \subseteq \sigma$ is non-empty, then $\tau \in K$.

The sets $\sigma \in K$ are called **simplices** or faces. The **dimension** of a simplex σ is defined as $|\sigma| - 1$. The dimension of the complex K is the maximum dimension of any of its simplices.